THE HENSELIZATION OF A LOCAL RING IS A FILTERED COLIMIT OVER ELEMENTARY ETALE NEIGHBOURHOODS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10215-10224 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10226-10240, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Henselization of a local ring and elementary etale neighbourhoods; the identification is retained.

Source reference. Original lemma 04GV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10240–10240. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $s \in S$. Then we have*

$$\mathcal{O}_{S,s}^h = \operatorname{colim}_{(U,u)} \mathcal{O}(U)$$

where the colimit is over the filtered category which is opposite to the category of elementary étale neighbourhoods (U, u) of (S, s).

Proof. Let $\operatorname{Spec}(A) \subset S$ be an affine neighbourhood of s. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to s. With these choices we have canonical isomorphisms $\mathcal{O}_{S,s} = A_{\mathfrak{p}}$ and $\kappa(s) = \kappa(\mathfrak{p})$. A cofinal system of elementary étale neighbourhoods is given by those elementary étale neighbourhoods (U, u) such that U is affine and $U \to S$ factors through $\operatorname{Spec}(A)$. In other words, we see that the right hand side is equal to $\operatorname{colim}_{(B,\mathfrak{q})} B$ where the colimit is over étale A-algebras B endowed with a prime $\mathfrak{q}$ lying over $\mathfrak{p}$ with $\kappa(\mathfrak{p}) = \kappa(\mathfrak{q})$. Thus the lemma follows from Algebra, Lemma 04GV. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.